

A Packing of 10 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.962710. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.933072, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.962710 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 10$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.933072, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 10/9 = 2.962963$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N10.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
987570097641769754207370807709506769128102560722073143041427235510768873
476556916641413070136683781716759636521425459966048094873881454158843413
456973096527111263072281623070759926401051778695
```

denominator:

```
333333333332643979119711347401954185872867801765748445597501362258660313
033750094568708050605551082775529464812467302972345230971059090023185007
801924139790561058613903575230925709702888066556
```

B Coordinates

10 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N10.txt`.

```
-378860603191 -53568199433 -296525431898 -10914594546809447982 -483996986855 108194033353 3358692010155655174798 -336821704603 397703261571 -59811842416
112156492382 195607628266 25526282652 594409726100 -163919691330 382909407164 171385633295 48821789276281659838173 187887940509 317648049887 -88799911363
-499648153957 -207658361835 246429543089 29016624127369641697623 136882705043 -140900221238 -52636971628851489446 51840157979 288761077620 120505310303
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```